C3-3

1. 若$\hat{P}$和$\hat{Q}$具有同一组完备的共同本征函数$\{\psi_n\}$

则 $\hat{P}\psi_n = p\psi_n$，$\hat{Q}\psi_n = q\psi_n$

任意函数 $\psi = \sum_n c_n\psi_n$

$$[\hat{P},\hat{Q}]\psi = (\hat{P}\hat{Q}-\hat{Q}\hat{P})\psi$$
$$= \hat{P}\hat{Q}\sum_n c_n\psi_n - \hat{Q}\hat{P}\sum_n c_n\psi_n$$
$$= \hat{P}\sum_n c_n q\psi_n - \hat{Q}\sum_n c_n p\psi_n$$
$$= \sum_n c_n q p\psi_n - \sum_n c_n p q\psi_n = 0$$

则$[\hat{P},\hat{Q}]=0$，$\hat{P}$和$\hat{Q}$一定对易.

∴两不对易的算符，不可能拥有一组完备的共同本征函数. 得证.

2. $\hat{D}_x(a)\psi(x) = \psi(x-a)$

$\hat{f}(x)$、$\hat{D}_x(a)$对易，则 $\hat{f}(x)\hat{D}_x(a) = \hat{D}_x(a)\hat{f}(x)$

$$\hat{f}(x)\hat{D}_x(a)\psi(x) = \hat{f}(x)\psi(x-a)$$

又 $\hat{f}(x)\hat{D}_x(a)\psi(x) = \hat{D}_x(a)\hat{f}(x)\psi(x) = \hat{f}(x-a)\psi(x-a)$

则 $\hat{f}(x)\psi(x-a) = \hat{f}(x-a)\psi(x-a)$

∴ $\hat{f}(x) = \hat{f}(x-a)$，得证